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Deep Neural Networks - Lecture 4

Marcin Mucha

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Plan

Saturation

Batch Normalization

Skip connections

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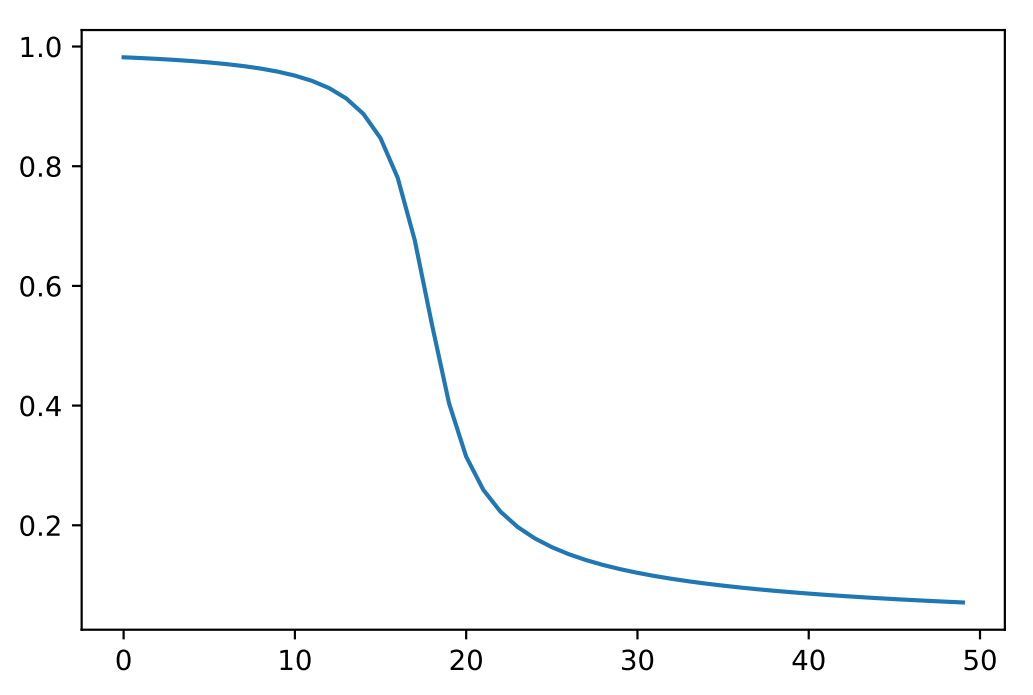
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Motivation

Consider a single sigmoid unit trying to learn on a single example $x = 1, y = 0$. The initial weight is $w = 2.0$ and bias is $b = 2.0$ (initial prediction is $\frac{1}{1+e^{-4}} \approx 0.98$). We use a quadratic loss function. This is how loss behaves over 50 epochs.

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This unit is a *saturated* unit.

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$$\frac{\partial L(w, b)}{\partial w} = 2\sigma(w + b) \cdot \sigma(w + b) (1 - \sigma(w + b)) .$$

The second term makes this really small when prediction is close to one - very counter-intuitive.

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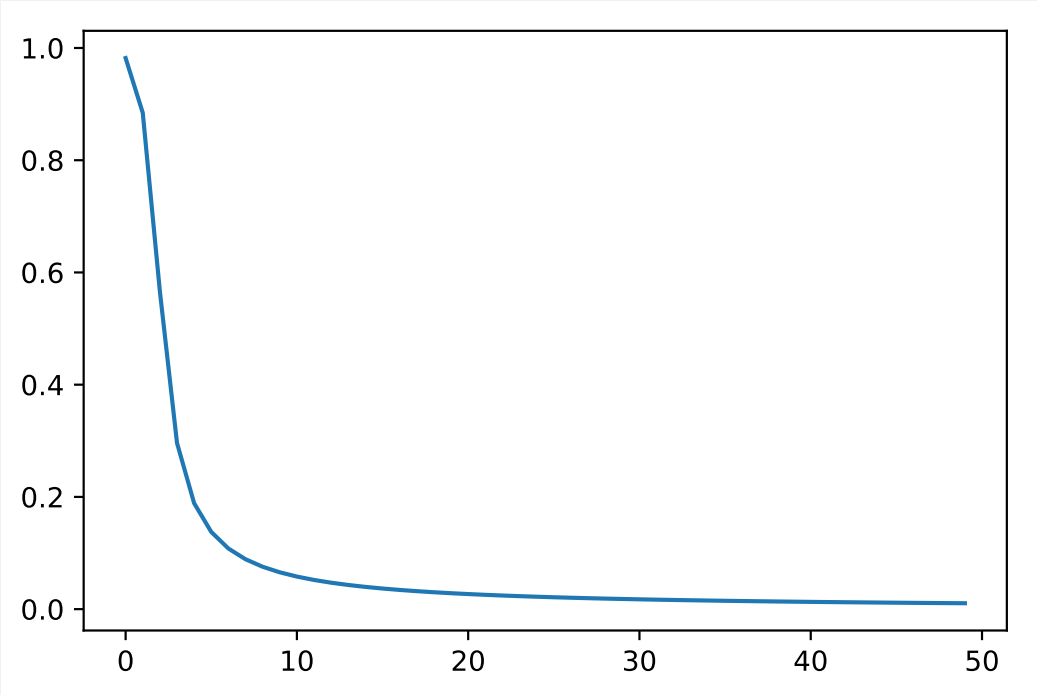
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This is perfect!

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Sigmoid units can learn in a very strange way when using GD. But sometimes you can help it.

When using sigmoid activations in the output layer, square loss is often a bad idea. Note: loss is often dictated by other considerations.

Real message: when facing difficulties, think about what GD is really doing and figure out the problem!

Exercise

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Answer: Of course it can. You can control the initial values in f^l as follows:

- Set biases to 0.
- Rescale inputs so that they are zero-centered and have unit variance.
- Set initial weights to small random values. How small?

Exercise, ctd

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A related approximate analysis balances forward activation and backward gradient scales, leading to so-called Glorot weights with standard deviation of $\sqrt{\frac{2}{m_i + m_{i+1}}}$).

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Exercise: Why it makes less sense to normalize g^l? Give a handwaving argument.

Answer: Outputs of g^l can have rather complicated distributions. Scaling and shifting might not guarantee stable distributions. That said, recent papers actually advocate this. Apparently it often actually works better!

Implementation

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No longer clear what is really happening inside.
Instead...

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$$v_i^l = \frac{1}{s}\sum_{j=1}^{s}(f_{i,j}^l - \mu_i^l)^2.$$

This is a vector of mini-batch estimates of variances of each f_i^l.

Solution, ctd.

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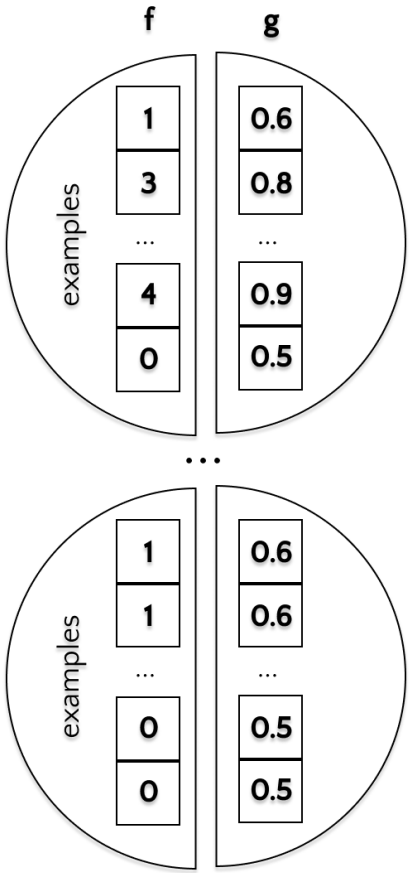
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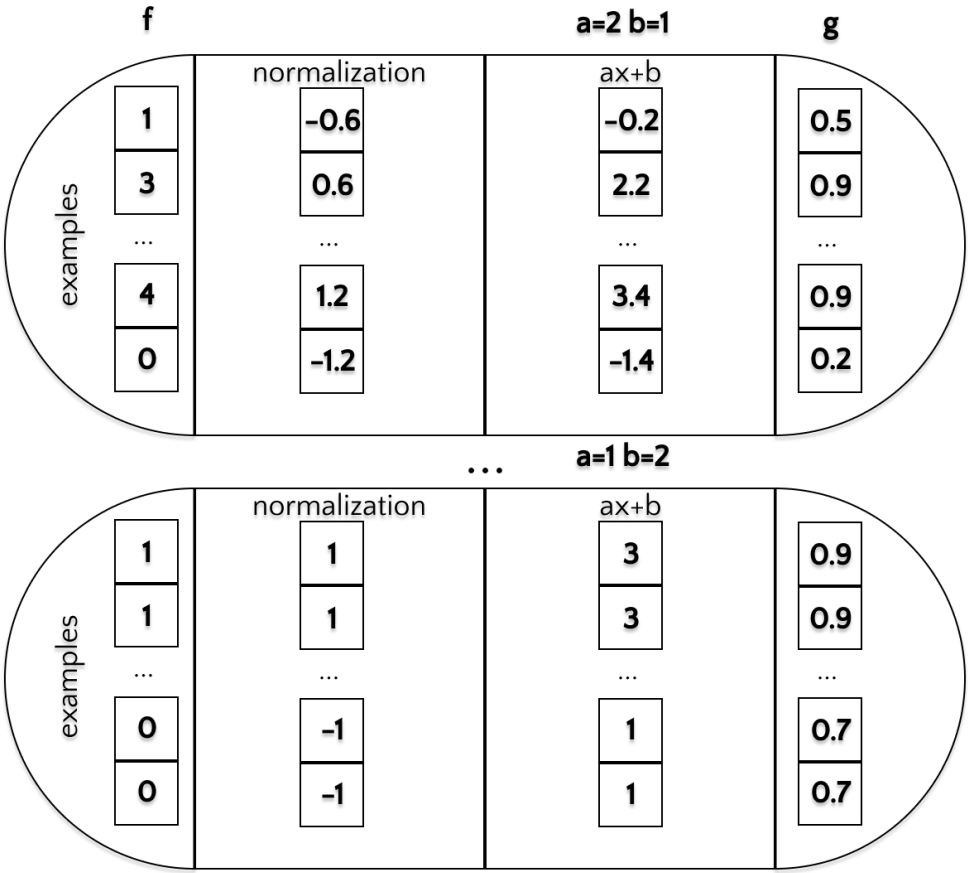
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This is a significantly more complicated computation: each $BN^l_{i,j}$ depends on all $f^l_{i,j}$. But can still compute gradients of L over g^l, BN^l and f^l, and consequently over w's, b's, γ's and β's.

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Extra details

When using batch normalization, we do not need biases, they get removed anyway. β's acts as biases.

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During training, use mini-batch statistics and maintain running estimates of the mean and variance. During evaluation, use the frozen running estimates from training; the learned γ and β stay unchanged.

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- Reduces saturation for sigmoid and tanh units and deactivation of ReLU units. This makes training deeper networks possible.
- No careful initialization required.

Recent research I

Laarhoven [2017] argues that:

- If using batch normalization, using L2 regularization as well might be crucial for convergence, BUT
- The L2 regularization coefficient only influences effective learning rate!

Recent research II

Later, Santurkar et al. [2018] argue that:

- Batch normalization helping with internal covariate shift is a myth.
- What batch normalization really does: it smoothens the loss funcion and its gradient.

Plan

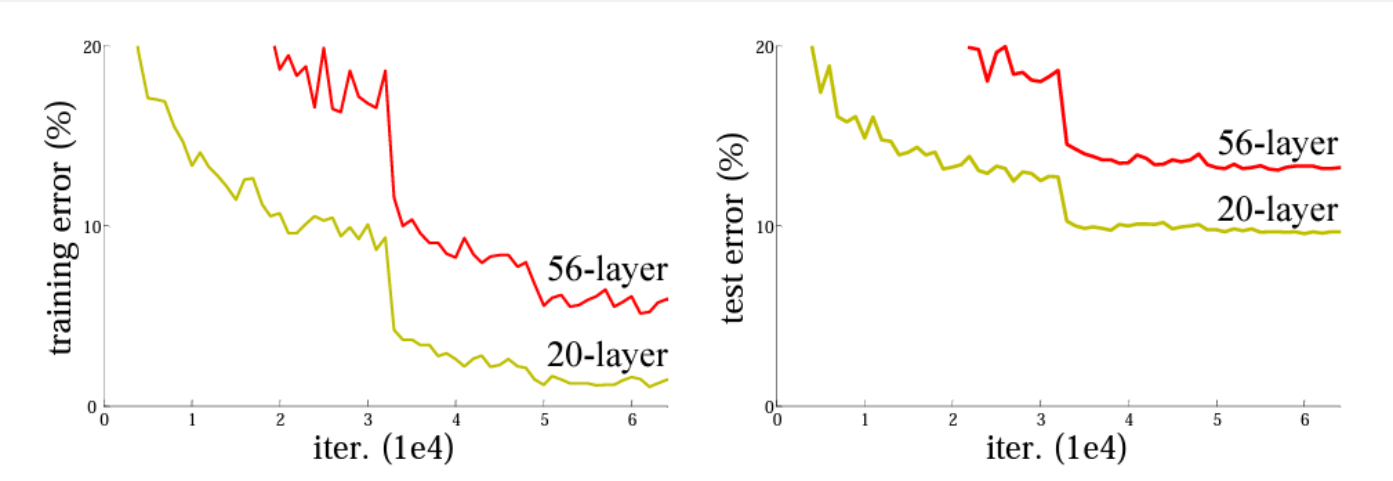
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Deeper networks

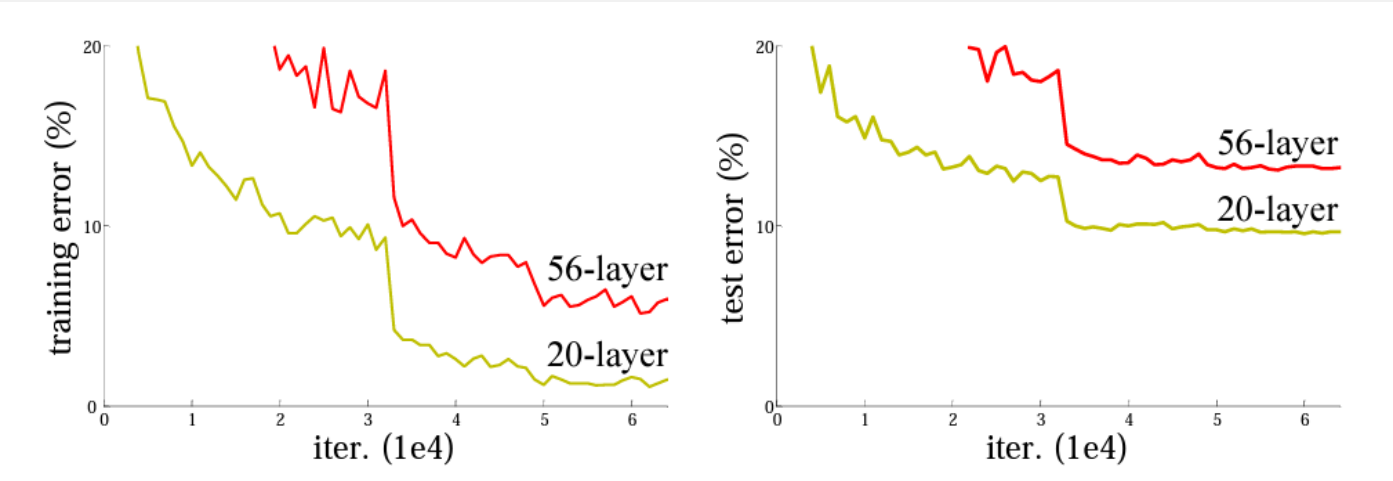
- Deeper network should in theory work better (on the train set) since it can just use identity mapping on extra layers.
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What seems to be the problem?

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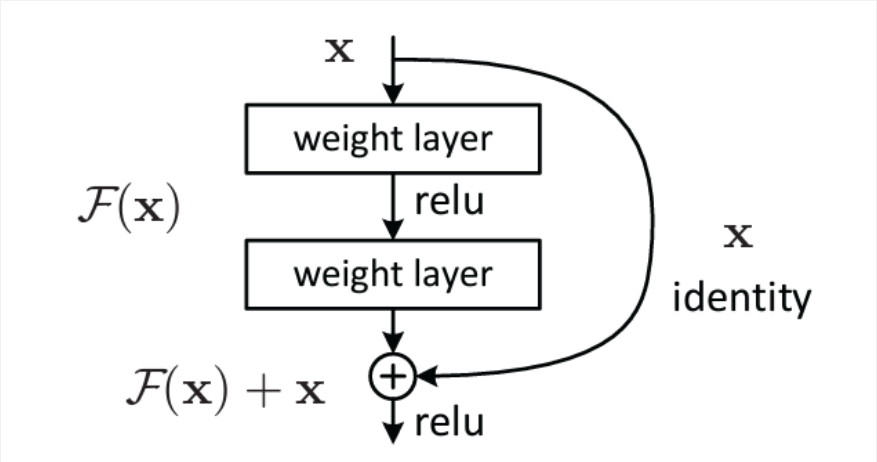
What seems to be the problem?

Identity seems to be a rather difficult mapping to learn!

Residual connections

Solution: residual connection.

- Signal is copied several layers forward.
- It is ADDED to layers' output, so it learns the RESIDUAL.
- To go deeper we learn the 0 mapping, not identity - easier.
- Dimension needs to agree.
- No new parameters!

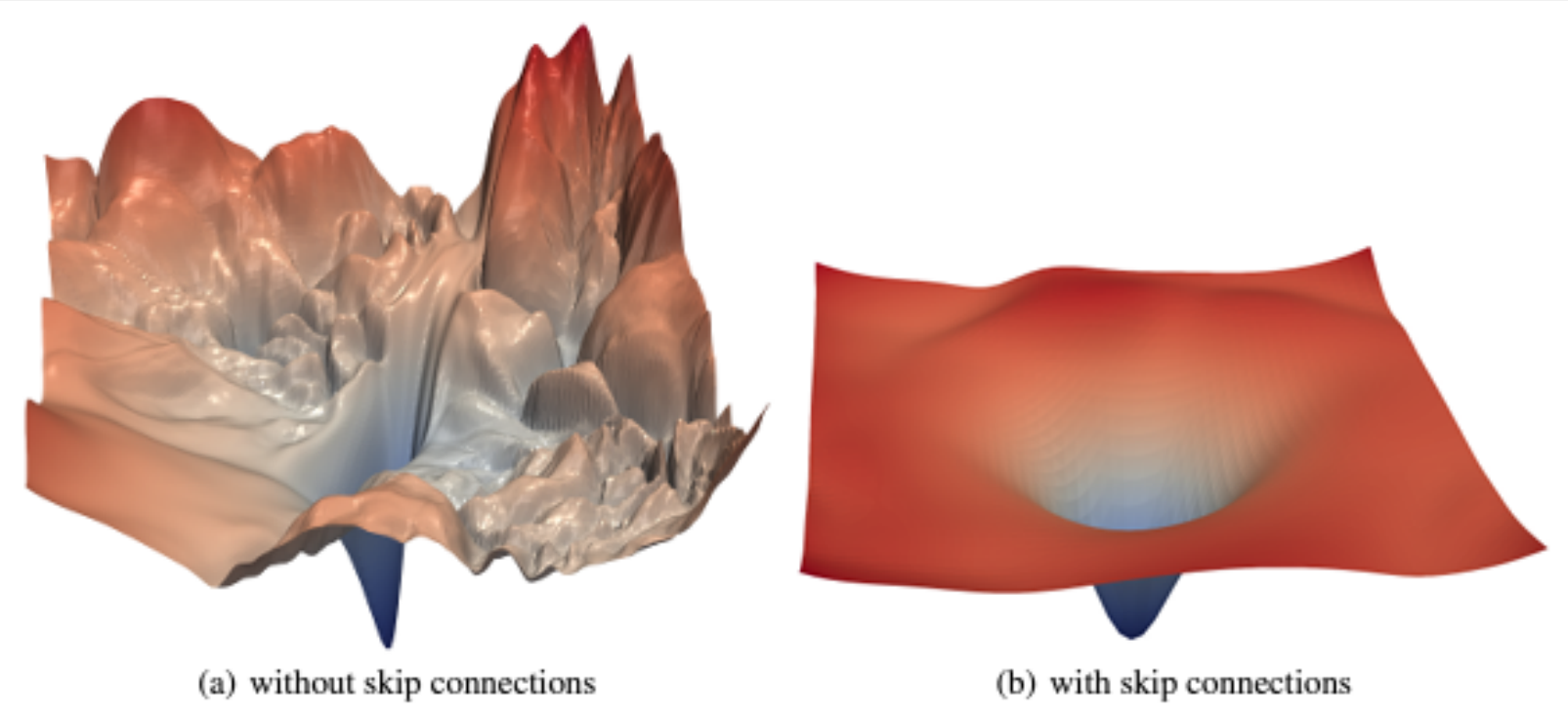


Does it work?

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Spectacular success stories, e.g. ResNet (next lecture) or transformers.

Today: a suggestive image (for ResNet).



(a) without skip connections

(b) with skip connections

Concatenation

Sometimes it makes more sense to use a skip connection to APPEND an early layer to deeper layer.

Useful for "annotating the input" style problems with dense outputs, like image segmentation, etc. (see U-Net in next lecture). Even for shallow networks

Typically these are called "skip connections" (vs "residual connections"), but not always!

Extra reading

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Original batch norm paper

Batch norm with L2 regularization

Exploration of what batch norm does

Original ResNet paper

Visualization of skip connection landscape





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